NITHYASRI SRIKUMAR
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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science — **Philadelphia, PA**
Master of Science in Engineering in Computer and Information Science — Expected May 2027

SRM Institute of Science and Technology, College of Engineering — **Chennai, India**
Bachelor of Technology in Computer Science and Engineering — May 2023
Cumulative GPA: 3.98/4.00 (scale: 0.00-4.00)

SKILLS

Machine Learning for Robotics: Reinforcement learning (offline RL, CQL, flow Q-learning, PPO, SAC, TD3), imitation learning and behavioral cloning, VLA models, flow- and diffusion-based policies, sim-to-real transfer
Robotics Software: ROS 2 (Jazzy), pure pursuit path tracking, SLAM, particle filter localization, MPC, MPPI
Programming: Python, C++, C, Java, JavaScript
ML & Simulation Frameworks: PyTorch, JAX, Isaac Lab, Isaac Sim, RSL-RL, NumPy, OpenCV, W&B, Hugging Face Transformers, FAISS

RESEARCH EXPERIENCE

Research Assistant, TrustML Group, University of Pennsylvania, Jan 2026 – Present — **Philadelphia, PA**
Advisors: Dr. Osbert Bastani, Dr. Dinesh Jayaraman
Co-first-author paper under review at ICLR 2027

- Developed QGRQL, an offline RL method for flow-matching policies built on RQL, that selects actions with Q-guided targets and a co-trained critic. On the same OGBench task set, it reaches a 67.3% average success rate, versus 64.1% for RQL and 37.9% for FloQ, and also beats FQL.
- Diagnosed value-function failures in offline training (out-of-distribution overestimation, value divergence, target-network corruption) by building diagnostic instrumentation in JAX.
- Implemented a trust-region mechanism with projection and n-step Bellman backups for action-chunked policies; ablations over chunk horizon and the step at which the critic gradient is introduced showed that optimizing action targets from the first iteration worked best.
- Earlier: built and evaluated diffusion policy (DDPM) and DDPG+BC pipelines on self-collected demonstrations for online RL in Isaac Lab.

PROFESSIONAL EXPERIENCE

Wells Fargo, Software Engineer, Aug 2023 – Jul 2025 — **Bengaluru, India**

- Designed and implemented RESTful APIs and modular Spring Boot microservices to modernize legacy mainframe systems into scalable, cloud-ready applications, decommissioning three monolithic services and reducing infrastructure costs by 20%.
- Architected backend service layers with role-based access control (OAuth2, JWT, SSO), improving system reliability and reducing auth-related defects by 40%.
- Refactored business logic into modular, testable service components, improving production stability and reducing incident rates by 25%.
- Collaborated across frontend (React.JS) and backend (Spring) layers to build end-to-end distributed systems with clean interface contracts and performance optimization.

PROJECTS

Autonomous Drone Racing with RL and Sim-to-Real Transfer, ***ESE 6510, University of Pennsylvania,*** Jan 2026 – May 2026

- Developed end-to-end RL-based autonomous drone racing policies for a quadrotor in NVIDIA Isaac Lab, spanning large-scale simulation training and real-world deployment via a ROS 2 pipeline; completed three racing laps in 17.14 s.
- Trained powerloop racing policies using PPO with domain randomization over aerodynamic drag, PID gains, and thrust-to-weight ratio, achieving stable sim-to-real transfer.
- Designed a gate-centric vertical progress reward to overcome altitude local optima on a 7-gate track with 2.0 m obstacles, enabling consistent multi-gate traversal.
- Analyzed ROS 2 bag data from real-world flights to diagnose sim-to-real failure modes, including perception–control latency mismatches and policy instability to guide debugging.

F1TENTH RoboRacer: Autonomous Racing, ESE 6150, University of Pennsylvania, Aug 2026 – Present

- Implemented pure pursuit in C++ on ROS 2 (Jazzy), with waypoints from SLAM, particle filter localization, minimum-curvature and minimum-time racelines, and a velocity-dependent dynamic lookahead; averaged 8.7 s per lap on a ~20 m track on the real 1/10-scale car and placed 3rd in the class in Race 1.
- Completed reactive navigation labs (wall following, follow-the-gap); implementing LQR, MPC and MPPI for Race 2 and the final race.

PennOS - Operating System, ***Team of 3*****,** Sep 2025 – Dec 2025

- Implemented a Unix-like kernel in C with PCB-based process management and 4-level priority scheduling (foreground/background), enabling reliable context switching and full lifecycle handling (running, blocked, zombie, orphan).
- Developed a FAT-based file system and user-level threading subsystem with mutex-based synchronization, ensuring correct concurrent execution and resource cleanup across processes.

Retrieval-Augmented QA System with Transformer Models, ***Independent Project*****,** Jun 2024 – Aug 2024

- Built RAG-based QA system combining Sentence Transformers + FAISS retrieval with DeBERTa extractive QA, enabling scalable and context-aware question answering over large corpora.